LITERATURE SURVEY TABLE

Research Methodology in Computer Science

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Section: B1[CCVT] **Date:** 15/9/2026

Assignment Task
Select one research topic and find **7 relevant research papers**. Read the papers and prepare a Literature Survey Table.
Use **short points or phrases** in the table and write the information in your **own words**.

How to Fill the Literature Survey Table

1. **Search:** Find relevant papers using IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, Scopus or Google Scholar.
2. **Citation:** Write the author(s) and publication year.
3. **Objective / Problem:** Write what problem the researchers are trying to solve or study.
4. **Methodology:** Write the method, algorithm, model, experiment, dataset or technique used.
5. **Key Finding:** Write the main result or contribution of the paper.
6. **Limitation:** Write an important limitation, unanswered question or future research area mentioned in the paper.

Research Topic: Machine Learning for Network Intrusion Detection and False-Positive Reduction

References / Citations

Write the complete reference of the seven papers used above. Include author(s), year, paper title, journal/conference, and DOI or link where available.

1. [1] A. L. Buczak and E. Guven, "A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection," IEEE Commun. Surveys Tuts., vol. 18, no. 2, pp. 1153-1176, 2016, doi: 10.1109/COMST.2015.2494502.
2. [2] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "A Detailed Analysis of the KDD CUP 99 Data Set," in Proc. IEEE CISDA, 2009, pp. 1-6, doi: 10.1109/CISDA.2009.5356528.
3. [3] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," in Proc. ICISSP, 2018, pp. 108-116, doi: 10.5220/0006639801080116.
4. [4] R. Vinayakumar, M. Alazab, K. P. Soman, P. Poornachandran, A. Al-Nemrat, and S. Venkatraman, "Deep Learning Approach for Intelligent Intrusion Detection System," IEEE Access, vol. 7, pp. 41525-41550, 2019, doi: 10.1109/ACCESS.2019.2895334.
5. [5] C. Yin, Y. Zhu, J. Fei, and X. He, "A Deep Learning Approach for Intrusion Detection Using Recurrent Neural Networks," IEEE Access, vol. 5, pp. 21954-21961, 2017, doi: 10.1109/ACCESS.2017.2762418.
6. [6] B. A. Tama, M. Comuzzi, and K.-H. Rhee, "TSE-IDS: A Two-Stage Classifier Ensemble for Intelligent Anomaly-Based Intrusion Detection System," IEEE Access, vol. 7, pp. 94497-94507, 2019, doi: 10.1109/ACCESS.2019.2928048.
7. [7] M. A. Ambusaidi, X. He, P. Nanda, and Z. Tan, "Building an Intrusion Detection System Using a Filter-Based Feature Selection Algorithm," IEEE Trans. Comput., vol. 65, no. 10, pp. 2986-2998, 2016, doi: 10.1109/TC.2016.2519914.

No.	Citation	Objective / Problem	Methodology	Key Finding	Limitation
1	Buczak & Guven (2016)	Survey ML/data-mining methods for cyber intrusion detection.	Review of ML techniques (SVM, trees, clustering); compares studies and datasets.	Maps method strengths and gaps; stresses labeled-data and real-time needs.	Survey only; notes lack of current datasets and high false-alarm rates.
2	Tavallaee et al. (2009)	Expose flaws in the KDD'99 IDS benchmark.	Statistical analysis of KDD'99; propose refined NSL-KDD; test classifiers.	Redundant records bias results; NSL-KDD gives fairer evaluation.	Still synthetic, dated traffic; not fully like modern networks.
3	Sharafaldin et al. (2018)	Create a realistic, labeled modern IDS dataset.	Built CICIDS2017 with real attacks; extracted flow features; tested ML models.	Provides up-to-date benchmark; finds informative flow features.	Class imbalance; single testbed may limit generalization.
4	Vinayakumar et al. (2019)	Scalable deep-learning IDS for host and network.	Deep neural nets vs classical ML on KDD, NSL-KDD, UNSW, CICIDS.	DNNs generalize across datasets with competitive detection.	High compute/training cost; tuning needed; false alarms remain.
5	Yin et al. (2017)	Improve IDS accuracy with deep sequence models.	Recurrent neural network (RNN-IDS) on NSL-KDD; binary and multiclass.	RNN beats traditional ML in detection accuracy.	Longer training; evaluated mainly on NSL-KDD.
6	Tama et al. (2019)	Boost anomaly detection and cut false alarms.	Two-stage classifier ensemble + hybrid feature selection; NSL-KDD & UNSW-NB15.	Higher detection and accuracy than single models.	Added model complexity; only two datasets tested.
7	Ambusaidi et al. (2016)	Select informative features to improve IDS efficiency.	Filter-based mutual-information feature selection with LS-SVM classifier.	Fewer features give faster, accurate detection.	Older datasets (KDD'99/NSL-KDD); filter ignores feature interaction.

Before Submission:

☑ 7 relevant papers selected ☑ Citations recorded

☑ Table completed ☑ Information written in own words ☑ References / links provided

Search → Read → Cite → Fill the Table